

Question number	Answer	Additional guidance	Mark
3(a)(i)	An explanation that includes four of the following points: • for equilibration (1) • because mixing the solutions at different temperatures would make the temperature $\neq 40\text{ }^{\circ}\text{C}$ (1) • (40 $^{\circ}\text{C}$ could be) optimum for urease (1) • (so if the mixture) {cooled down / warmed up} rate would change during experiment (1) • ensures only pH is varied (1)	Accept when mixing the temperature stays at 40$^{\circ}\text{C}$ / constant	(4)

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3(a)(ii)	An explanation that includes the following points: - run the experiment with a range of different concentrations (1) - choose a combination in which the rate of change is not too {high / fast} / too {low / slow} / time taken is not too {long / short} (1)		(2)

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3(b)(i)	A table showing the following features: - suitable table drawn (1) - headings of pH and concentration of ammonia (1) - with units (1) - data correctly entered (1)	Example of table drawn <table><tr><th>pH</th><th>concentration of ammonia / a.u.</th></tr><tr><td>3.0</td><td>0</td></tr><tr><td>4.0</td><td>10</td></tr><tr><td>6.5</td><td>60</td></tr><tr><td>6.8</td><td>80</td></tr><tr><td>7.3</td><td>100</td></tr><tr><td>8.0</td><td>60</td></tr><tr><td>9.0</td><td>30</td></tr></table>	pH	concentration of ammonia / a.u.	3.0	0	4.0	10	6.5	60	6.8	80	7.3	100	8.0	60	9.0	30	(4)
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3(b)(ii)	- figures read from graph and converted into rate (1) - appropriate units given (1)	Example of calculation: $60 \times 4 = 240 \text{ OR } 60 \div 15 = 4 \text{ OR}$ $60 \div (15 \times 60) = 0.07/0.067/0.0667$ (240) a.u. hour⁻¹ OR (4) a.u. min⁻¹ OR (0.07/0.067/0.0667) a.u s⁻¹	(2)

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3(b)(iii)	An answer that includes the following points: Similarities - both sets of results have a similar pattern (1) - both peak at same pH (1) Differences - no ammonia produced at pH 4 after 15 minutes but {some / 10 a.u.} produced at 60 minutes (1) - more ammonia produced at every pH after 60 minutes than after 15 minutes (1)	Accept 60 min graph starts at pH 3 and 15 at pH 4	(3)

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3(c)	0.36 a.u.min⁻¹ (1)	Accept answers in range of 0.3 to 0.4 a.u.min⁻¹ Accept other values with corresponding appropriate units e.g. rate expressed per second 0.003 to 0.007 a.u.sec⁻¹	(1)